

DeepSeek-V2: Efficient MoE Language Model with Multi-Head Latent Attention

ML researchers, NLP engineers, and graduate students

Total / Activated Params

236B / 21B

Context Length

128K tokens

Core Innovations

MLA + DeepSeekMoE

Positioning

Strong, economical, efficient

The LLM Scaling Dilemma: Performance vs. Cost/Latency

As models improve capability, but training cost rises sharply.
Memory scales with context length via KV cache.
Inference becomes bottlenecked by cache footprint and bandwidth.
Architectures that preserve quality while reducing active computation.

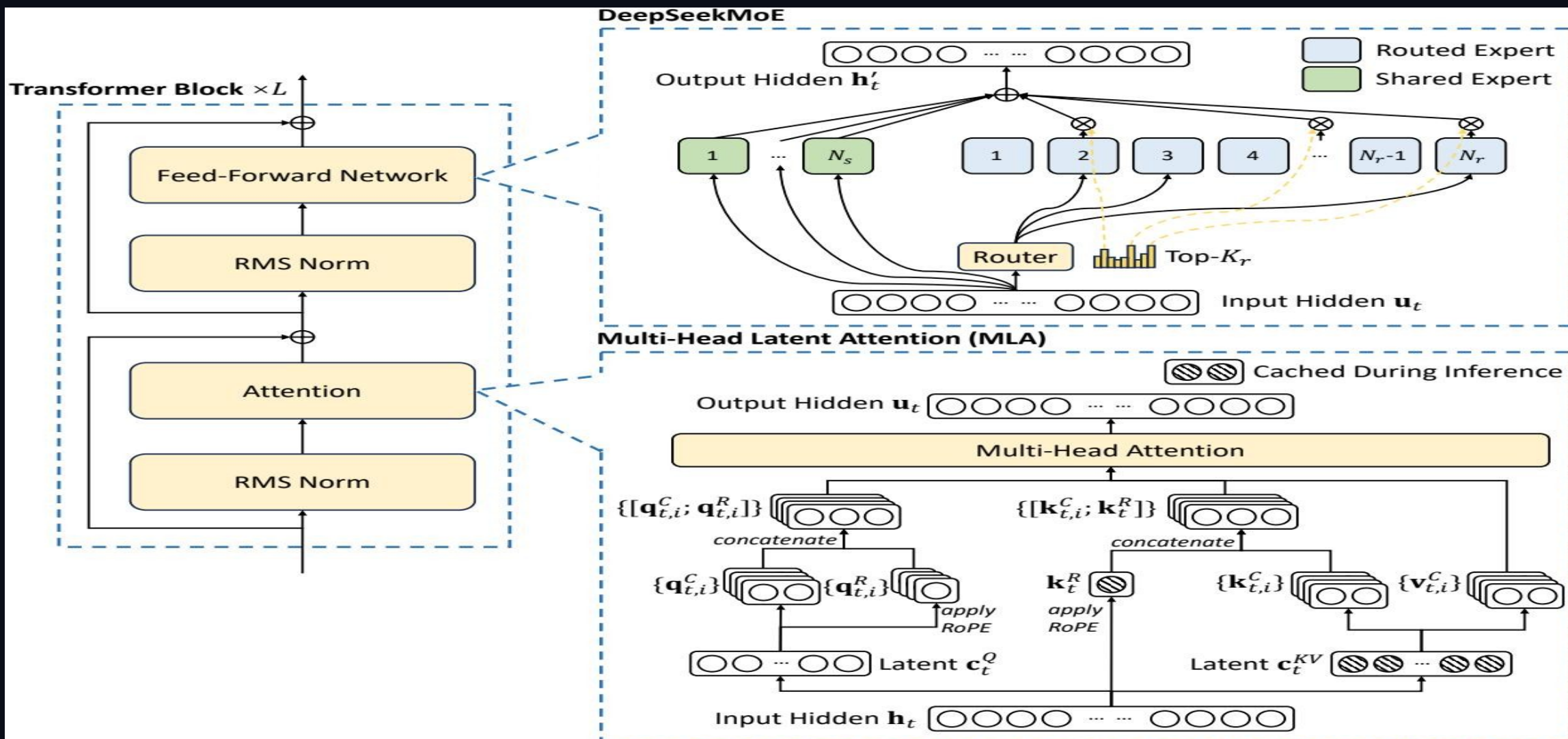
DeepSeek-V2 hypothesis

Use MLA to shrink attention-state memory.

Use DeepSeekMoE to scale total capacity economically.

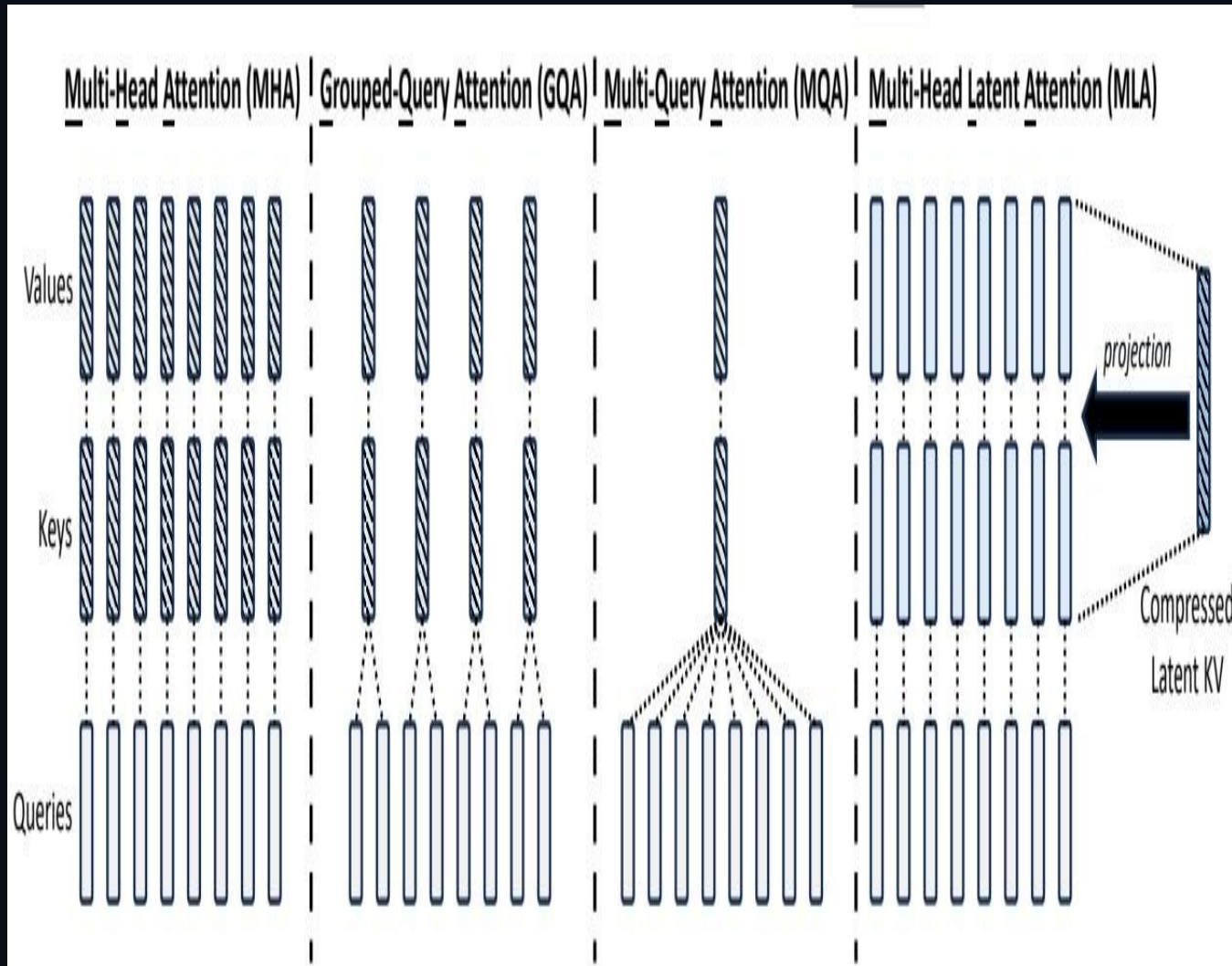
Target Pareto gains: quality, training cost, and throughput.

DeepSeek-V2 Architecture: MLA + DeepSeekMoE



Transformer block uses MLA in attention and DeepSeekMoE in FFN pathway.

Multi-Head Latent Attention (MLA): 93.3% KV Cache Reduction



Low-rank joint compression of keys/values into latent states.

Decoupled RoPE preserves positional modeling under compression.

Compared with MHA/GQA/MQA, MLA keeps quality while minimizing cache.

Reported KV cache reduction: 93.3%.

DeepSeekMoE: Fine-Grained Experts + Shared Expert Isolation

Two expert types: shared experts + routed experts.
Top-K routing activates sparse capacity per token.
Device-limited routing reduces cross-device communication.
Auxiliary load-balancing losses mitigate expert collapse.
Bias-update strategy stabilizes expert utilization during training.

Outcome

236B total parameters with only 21B activated.
High model capacity without dense-model training cost.
Compatible with large-scale long-context deployment.

Efficiency Gains vs DeepSeek 67B

Relative metrics reported for DeepSeek-V2 (no fabricated absolutes)

Training Cost

DeepSeek 67B → DeepSeek-V2

-42.5%

lower than DeepSeek 67B

KV Cache Size

DeepSeek 67B → DeepSeek-V2

-93.3%

reduction via MLA

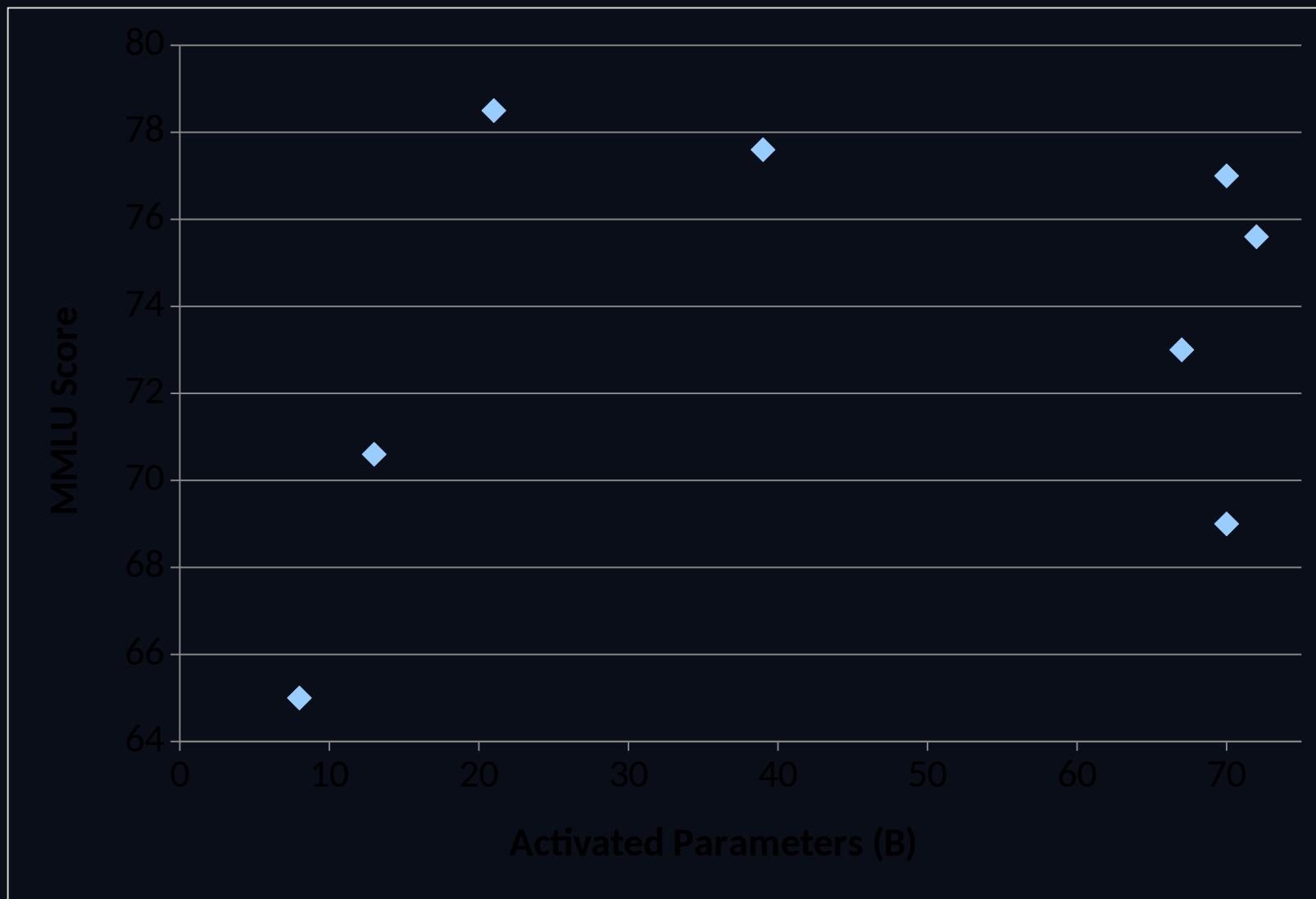
Generation Throughput

DeepSeek 67B → DeepSeek-V2

5.76×

higher max throughput

MMLU vs Activated Parameters: Pareto Position of DeepSeek-V2



Key points

DeepSeek-V2: (21B, 78.5)

LLaMA 3 70B: (70B, 77.0)

Mixtral 8x22B: (39B, 77.6)

Qwen1.5 72B: (72B, 75.6)

DeepSeek 67B: (67B, 73.0)

LLaMA 2 70B: (70B, 69.0)

Mixtral 8x7B: (13B, 70.6)

LLaMA 3 8B: (8B, 65.0)

DeepSeek-V2 delivers highest MMLU at much lower activated params than 70B-class dense models.

Alignment Results: DeepSeek-V2 Chat (RL)

SFT on 1.5M sessions + GRPO-based RL

AlpacaEval 2.0 (LC win rate)

38.9%

MT-Bench (overall)

8.97

AlignBench (overall, Chinese)

7.91

Reported to outperform all open-source and most closed-source models on Chinese alignment benchmarks.

Key Takeaways

- MLA is a practical attention upgrade: strong quality with 93.3% lower KV cache.
- DeepSeekMoE scales total capacity economically through sparse activation and expert design.
- DeepSeek-V2 reduces training cost by 42.5% vs DeepSeek 67B while improving capability.
- At 21B activated params, DeepSeek-V2 reaches top-tier open-source MMLU performance.
- 5.76× generation throughput gain enables more efficient real-world deployment.